CSM
Fall 2026

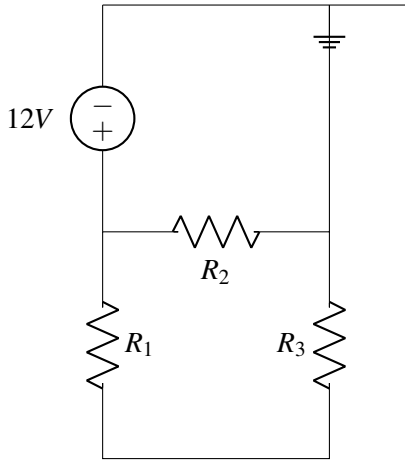
EE 64
Worksheet 2

1. Introduction to KVL, KCL

KVL: Kirchhoff's Voltage Law says that the sum of voltage drops going around a closed network (starting and ending at the same point) is 0

KCL: Kirchhoff's Current Law says that the sum of all currents flowing into a node are equal to the sum of all currents flowing out of a node. This is equivalent to saying that the sum of all currents flowing out of a node (representing flows in opposite direction as negative) is equal to the sum of all currents flowing into a node (again, representing flows in opposite direction as negative) which is equal to zero

Consider the following circuit, with $R_1 = 2\Omega$, $R_2 = 3\Omega$, and $R_3 = 4\Omega$:

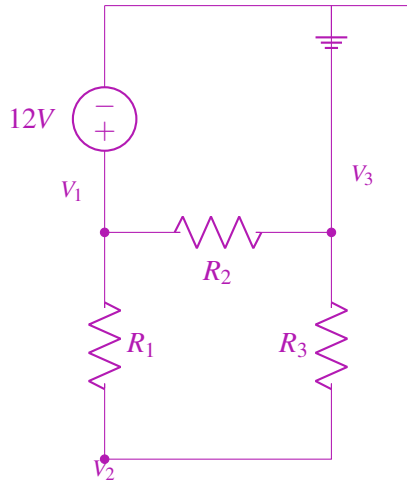


We will solve this picture using nothing but KCL and KVL in the following steps.

(a) First, draw a reference point (ground) on the circuit, and label all the nodes V_1, V_2, etc.

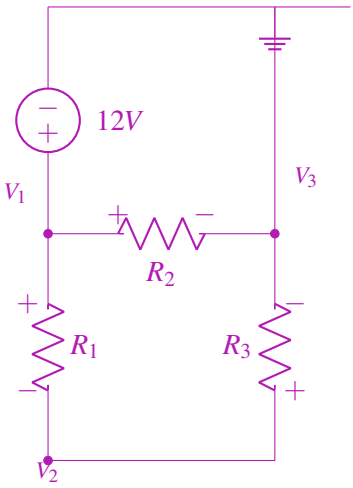
Solution: You should draw a ground on either side of the voltage source, and the 3 nodes (labelled any way) as shown below. We'll continue using this labeling convention for the rest of the problem.

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(b) Next, draw the +s and -s on the resistors.

Solution: While there is no incorrect way to do this (as long as you are consistent with the equations in future parts), the way of doing so that is consistent with the method used in this problem is:
For R_1, have the + be on the northside and the - on the southside.
For R_2, have the + be on the leftside and the - on the rightside.
For R_3, have the + be on the southside and the - on the northside.

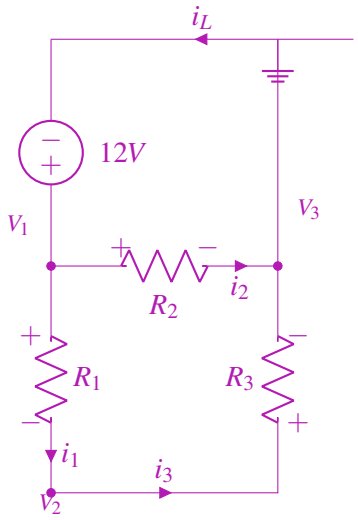


Meta: Make sure to go over what difference is made by the + and - location, namely the idea that Ohm's law is about ΔV, not just V.

(c) Finally, draw arrows indicating the direction of current, labeling them as I_1, I_2, etc.

Solution: Once again, as long as you are consistent for the next part, there isn't a wrong answer, but our equations use:
I_1 is an arrow going south between the source and V_1.
I_2 is an arrow going east between V_1 and V_3 around R_2.
I_3 is an arrow going east between V_2 and V_3 around R_3.
I_L is an arrow going west between ground and the negative terminal of the voltage source.
Our drawn on circuit should now look something like:

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(d) Now write all the equations given by Ohm's law, KCL and any voltages you can find relative to ground
Solution: Ohm's Law ($\Delta V = RI$):

$$V_1 - V_2 = 2I_1$$
$$V_2 - V_3 = 4I_3$$
$$V_1 - V_3 = 3I_2$$

We also see that V_1 is $12V$ above ground and V_3 is connected to the same node as the ground, giving us:

$$V_1 = 12V$$
$$V_3 = 0V$$

And here are our KCL equations:

$$I_1 = I_3$$
$$I_L = I_1 + I_2$$
$$I_L = I_2 + I_3$$

(e) Now that you have all equations written out, solve them to find all unknown voltages and currents.
Solution: We can start off by finding equivalent values. For simplicity, let's keep everything in terms of I_1. If we substitute all of these into the Ohm's Law formula, we have the current equations:
Now, we have the following values:

$$V_1 = 12V$$
$$V_2 = 8V$$
$$V_3 = 0V$$
$$I_1 = 2A$$
$$I_2 = 4A$$
$$I_3 = 2A$$

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(f) Now, we're going to solve the same circuit with Nodal Analysis. Notice that in the previous part, you had many equations are variables (corresponding to both voltage and current). This may make the system more tedious to solve. Instead, Nodal Analysis allows us to write a system of equations that only involves the node voltages.

Start by using the ground and voltage source to find any node voltages which are immediately known.

Solution: Because V_3 belongs to the same node as ground, we know $V_3 = 0V$. Also, the voltage drop across the source is $12V$, which equals $V_1 - 0$. Therefore, $V_1 - 12 = 0 \implies V_1 = 12$. So, we know that:

$$V_1 = 12V$$

$$V_3 = 0V$$

(g) Then, Ohm's law to write a KCL equation for each node purely in terms of voltages and resistances.

Solution: Applying KCL at the node V_2, we have

$$\frac{V_1 - V_2}{R_1} + \frac{V_3 - V_2}{R_3} = 0$$

Note that we have just written each of the incoming currents in terms of the voltage drop across the corresponding resistors. At this point, we actually have enough information to solve the equation, because V_1 and V_3 are known quantities. Plugging them in, we get

$$\frac{12 - V_2}{2} + \frac{0 - V_2}{4} = 0 \implies 6 - \frac{V_2}{2} - \frac{V_2}{4} = 0 \implies \frac{3}{4}V_2 = 6 \implies V_2 = 8V$$

Now that we've solved for all the node voltages, we have implicitly solved the entire circuit! Because we can always use the node voltages to find any currents we want. In particular,

$$I_1 = \frac{V_1 - V_2}{R_1} = \frac{12 - 8}{2} = 2A$$

$$I_2 = \frac{V_1 - V_3}{R_2} = \frac{12 - 0}{3} = 4A$$

$$I_3 = \frac{V_2 - V_3}{R_3} = \frac{8 - 0}{4} = 2A$$

And the current through the top loop is $I_2 + I_3 = 6A$ anticlockwise (applying KCL at the node V_3).

(h) When doing nodal analysis, would it have helped to write a KCL equation at nodes V_1 and V_3? Why or why not?

Solution: No, it would not have helped. We would have had to introduce an extra variable for the current through the top loop, and use this in each of our KCL equations. So it's ultimately the same problem, just with more equations and variables.

There is also the issue that we now have a variable for current which we cannot turn into a voltage (becuase there is no resistor in the top loop. This is not desirable, since in nodal analysis we want all our equations to be in terms of voltages. No currents anywhere!

It turns out that when doing in nodal analysis you can skip KCL for nodes which are directly adjacent to components like voltage sources. However, we don't lose any information, because we still incorporated the source voltage into the process of finding V_1 and V_3 in the beginning!

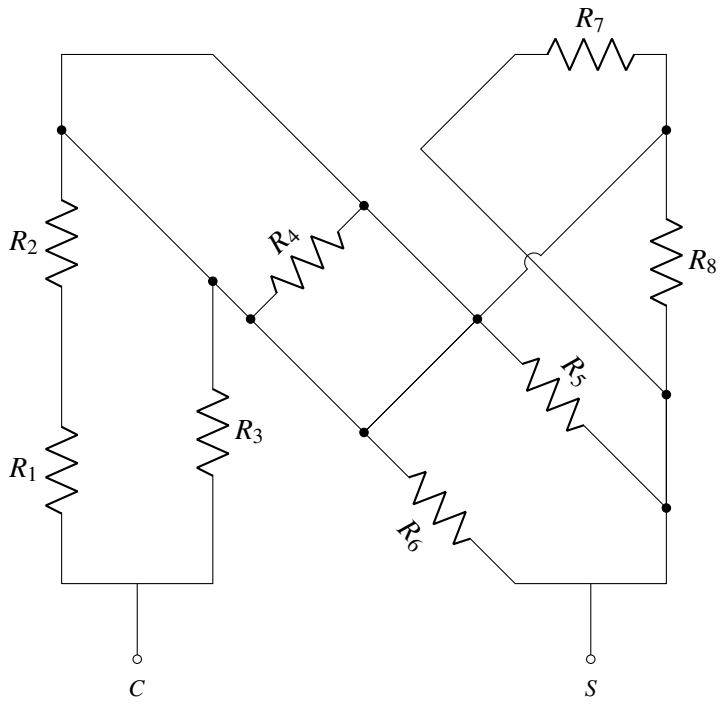
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2. A Resistor Network Problem

Learning Goal: The goal of this question is to understand equivalent resistance.

Meta:

(a) Calculate the equivalent resistance between the nodes C and S.



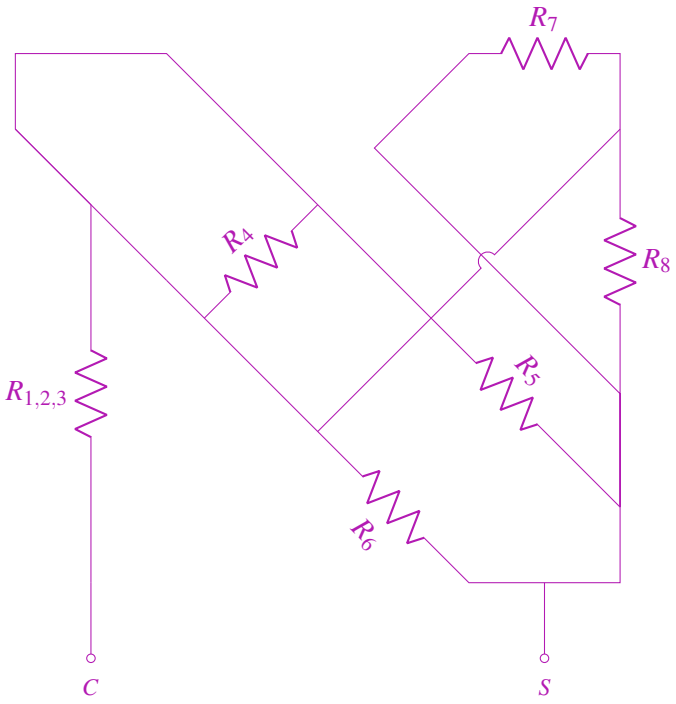
Solution: When we're given such a resistor network, it's helpful to identify what nodes the resistors share, to identify potential parallel and series configurations.

Let's begin at the C terminal: Resistors R_1 and R_2 share one end node (the node between the two resistors), but do not share any other nodes. We can consider R_1 and R_2 to be in a series configuration, meaning the equivalent resistance across them is $R_{1,2} = R_1 + R_2$.

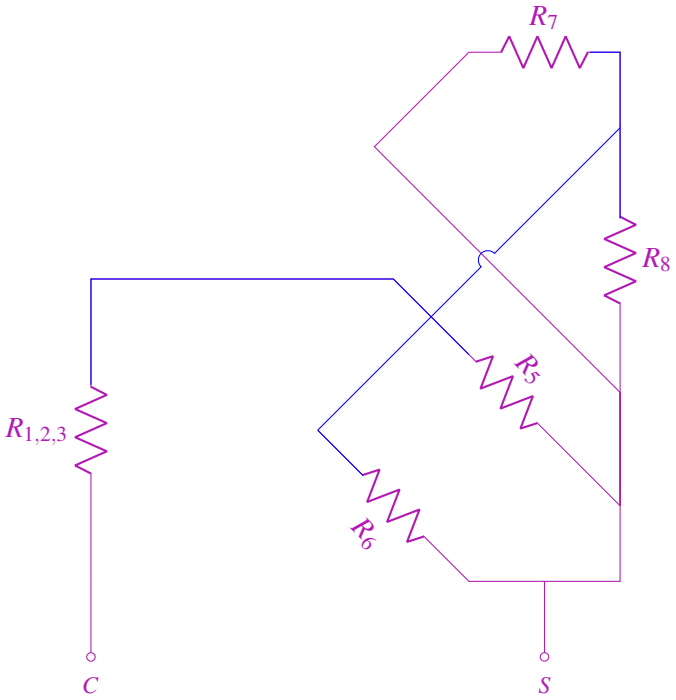
If we collapse this circuit, we see that $R_{1,2}$ is in parallel with R_3. Then, the equivalent resistance across R_1, R_2, and R_3 is $R_{1,2,3} = \frac{R_{1,2}R_3}{R_{1,2}+R_3} = \frac{R_1R_3+R_2R_3}{R_1+R_2+R_3}$.

Collapsing this left half of the resistive network, we should have an equivalent circuit as follows:

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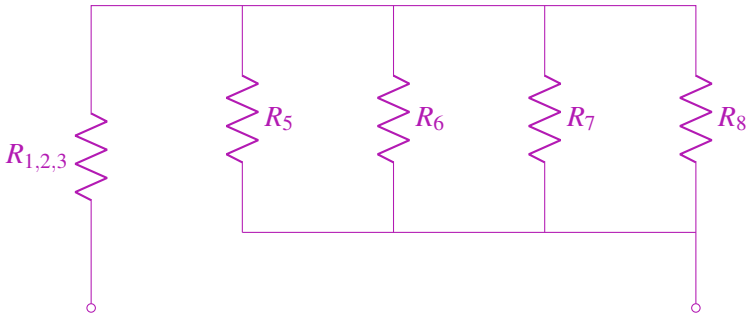


Now, notice that the R_4 resistor is shorted. Its two end nodes are the same node. We can then eliminate the R_4 resistor from consideration. We are then left with just the right side of the circuit. Let's try to simplify this circuit by analyzing the nodes:



Observe that resistors R_5, R_6, R_7, and R_8 all share two end nodes: one blue node (the same node as that on the right end of $R_{1,2,3}$, and the node that s belongs to. These four resistors are all in parallel. We can then simplify our circuit further to the following:

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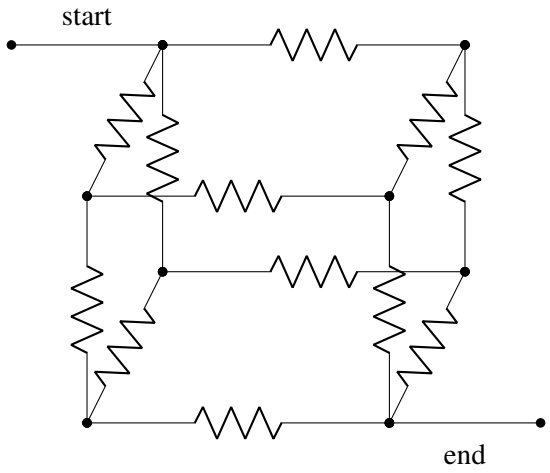
We can then use our series and parallel formulas to find the overall equivalent resistance between nodes C and S:

$R_{\text{parallel}} = (\frac{1}{R_5} + \frac{1}{R_6} + \frac{1}{R_7} + \frac{1}{R_8})^{-}1$

$R_{eq} = R_{1,2,3} + R_{\text{parallel}}$

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3. Rubik's Cube (but with resistors!)



In this question, we will guide you through how to apply nodes and resistance equivalence to solve a complex resistor network (like the cubic one above, assuming all resistors are identical with a resistance of R) and find the equivalent resistance between the top back left node and the bottom front right node.

Solution: For this question, encourage the students to follow through the questions and apply nodal analysis and equivalent resistance along the way. Assure them that with the symmetrical nature of this cubic network, the result will appear less complicated than they expect.

(a) As daunting as this question looks, fortunately we have the power of nodal analysis on our side!

i. First, label all the vertices of this cube with the letters A through H (starting with the top back left vertex as A and the bottom front right vertex as H).

Solution: Make sure to mention to the students that the starting node at the top back left should be labeled as A, while the ending node at the bottom front right should be labeled as H so that they can consistently follow the labeling convention in subsequent parts of the question.

ii. Next, group the vertices from B through H based on the **distances of the shortest paths** along the edges of the cube starting from A. For example, the distance of the shortest path from A to the bottom front left vertex will be 2 (assuming each edge of the cube is 1 unit long).

Solution: Make sure to clarify to the students that shortest path always starts from A (as our point of departure) and must go along the edges of the cube.

Solution: Starting from this part and extending to the rest of this entire question, we label all the vertices as follows:

- top back left - A
- top front left - B
- top front right - C
- top back right - D
- bottom back left - E
- bottom front left - F
- bottom back right - G
- bottom front right - H

(b) Now, consider the group of all the vertices v_i that are 1 unit away from A (the starting node in the top back left) and the resistors in and between these pairs of nodes (v_i, A). How are they related to each

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other (Hint: think in terms of equivalent resistance)?

Solution: For this part, encourage the students to think in terms of potential differences. Use the following scenario as an example to provide them with some helpful hints:
Suppose I have 2 pairs of nodes (A,B) and (C,D), both pairs have the same resistances between them $R_{A-B} = R_{C-D}$. What can we conclude about the potential differences (V_{A-B}, V_{C-D}) of both pairs?
Solution: Based on our labeling convention in the previous part, the nodes that are 1 unit away from A are B, D, and E. Since all these nodes are of the same distance away from the starting node A, and the resistance between these nodes and A is the same (R), the potential differences between A and these nodes are **all the same**! This means that the 3 resistors between A and the nodes that are 1 unit away from it are in parallel.

(c) Now, consider the group of all the vertices v_j that are 2 units away from A. Call it Group II. Let the group in the previous part be Group I. How many edges lead from a vertex in Group I to a vertex in Group II (how many resistors are involved in this case)?

Solution: To count how many edges lead from a vertex in Group I to a vertex in Group II, we can begin with any vertex from the set $\{B,D,E\}$ and count the number of distinct edges that lead from it 1 unit farther away from A. As we can see in the cube above, for each vertex v_i in Group I, there are exactly 2 distinct edges that lead to a vertex 1 unit farther away from A. So, there are $3 \times 2 = 6$ such edges in total. Since each edge contains a resistor, there are 6 resistors involved in total.

(d) How are Group I and Group II related with each other? Is there anything we can do to simplify the resistor network so far?

Solution: When teaching, encourage the students to think about the following questions before working on this part:

- What happens if you merge the vertices(nodes) with the same potentials into one?
- How can we apply this to Group I, Group II and the resistors involved in both groups?

Solution: When we have nodes with the same potential, if we merge them into one, the overall potential difference between the merged endpoint and A still remains the same. This implies that merging all nodes with the same potential into one junction won't affect the overall structure or resistance of this resistor network!
Since nodes in Group II are 1 unit away from the nodes in Group I, and similarly, all resistors in and between are in parallel as well, we can merge their start nodes and end nodes into a single junction as well! Therefore, the overall resistors between the vertices in Group I and A are in series with those between the vertices in Group I and the vertices in Group II. So essentially, we have 3 resistors (from Group I) in parallel, and overall in series with 6 resistors (between Group I and Group II) in parallel.

(e) Finally, How far away is H (the ending node) from A (the starting node)? How many edges are there that lead from the nodes in Group II to H? Using equivalent resistance, is there anyway we can simplify all the circuits along the edges from the nodes in Group II to H? (Hint: think in terms of symmetry and what you've found in part (b))

Solution: Here comes the symmetrical beauty that underlines this cubic network! If we rotate the entire cubic resistor network by 180 degrees about the space diagonal from A and H, we will exactly end up with A swapped with H. What this really means is that all the edges that lead from the vertices in Group II to H are equivalent to the edges that first extend 1 unit away from A in part (b)! This further implies that the 3 resistors along these edges are in parallel again (as we converge back onto one end node)!

(f) Combining all your findings together, what is the overall resistance between the starting node on the top back left and the ending node on the bottom front right in this cubic resistor network?

Solution: For the final part, encourage the students to try "flattening" the 3D resistor network onto

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a plane. Make sure to bring up how they can apply all the resistance equivalence they've discovered over the previous parts to map out all the resistors based on the nodes and their shortest distances from A.

Solution: Let's recap what we've known so far, go through the groups of nodes and the resulting junctions that we merged from the previous parts.

i. Starting from A, we have 3 resistors in parallel. Over a distance of 1 unit, we reach the nodes B, D, E in Group I. By the equivalent resistance in a parallel circuit, the overall resistance of these 3 resistors is $\frac{R}{3}$.

ii. Between Group I and Group II, since the distance in and between the vertices from the 2 groups is the same (= 1 unit), and each edge carries the same resistor, this means all edges share the same potentials, therefore, we can merge all the start nodes into one junction T_0, and all the end nodes into another junction T_1. T_0 will contain the vertices B, D, E, and T_1 will contain the vertices C, F, G. Overall between the merged junctions, we have 6 edges (or resistors) in parallel with each other. Therefore, the overall resistance of this part will be $\frac{R}{6}$

iii. Between Group II and the end node H, by symmetry from part(e), we can apply what we've done exactly for Group I and A. Hence, the overall resistance of this part will be again $\frac{R}{3}$.

iv. All these 3 parts are in series with each other (from A to H), hence, by the equivalent resistance of a series circuit, the total resistance between A and H is equal to

$$\frac{R}{3} + \frac{R}{6} + \frac{R}{3} = \frac{5}{6}R$$

Again, with the power of nodal analysis and equivalent resistance, anything is possible! □

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